

Unbiasedness of Advantage Inheritance in Recursive Language Models

Pablo Leyva

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Setup

Let π_θ be a policy with parameters $\theta \in \mathbb{R}^d$. Consider an RLM tree $\mathcal{T}_g$ rooted at root rollout g , with descendant nodes $v \in \mathcal{T}_g$ each carrying a token sequence y_v sampled from π_θ . Let $r_g \in \mathbb{R}$ be the reward at the root, computed solely from the root's **FINAL** answer. Define the GRPO advantage at the root as

$$A_g = \frac{r_g - \mu_r}{\sigma_r + \delta}, \quad \mu_r = \frac{1}{G} \sum_{g'=1}^G r_{g'}, \quad \sigma_r = \sqrt{\frac{1}{G} \sum_{g'=1}^G (r_{g'} - \mu_r)^2},$$

with $\delta > 0$ a small regulariser. The *advantage-inheritance estimator* at descendant v is

$$\hat{g}_v(\theta) = \nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v) A_g,$$

where ctx_v is the context (parent prompt + REPL state) seen by v .

Assumption 1 (Conditional independence of node score and reward residual). *For every descendant $v \in \mathcal{T}_g$,*

$$\mathbb{E}[\nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v) (r_g - \mathbb{E}[r_g \mid \text{ctx}_v])] = 0. \quad (1)$$

Equivalently, r_g given ctx_v has mean independent of small θ perturbations through y_v alone, modulo the path of ancestors that produced ctx_v .

Theorem

Theorem 2 (Unbiasedness of advantage inheritance). *Under Assumption 1, the per-node inheritance estimator is unbiased for the policy gradient contribution at v :*

$$\mathbb{E}[\hat{g}_v(\theta)] = \mathbb{E}[\nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v) r_g] \left(+ O(1/G) \right),$$

where the $O(1/G)$ term arises from the inclusion of r_g in μ_r, σ_r . Consequently the aggregate estimator over the whole tree, with the recursive $1/k_v$ weights $w_v = \prod_{u \in \text{ancestors}(v)} 1/k_u$, satisfies

$$\mathbb{E}\left[\sum_{v \in \mathcal{T}_g} w_v \hat{g}_v(\theta)\right] = \nabla_\theta J_{\mathcal{T}_g}(\theta) + O(1/G),$$

where $J_{\mathcal{T}_g}$ is the expected reward functional over trees rooted at g .

Proof

Proof. We first prove the per-node statement. Let $b := \mathbb{E}[r_g \mid \text{ctx}_v]$. By the score-function identity $\mathbb{E}[\nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v)] = 0$ when conditioning on ctx_v (a constant in this conditional expectation), we have for any constant c :

$$\mathbb{E}[\nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v) c \mid \text{ctx}_v] = 0.$$

In particular $\mathbb{E}[\nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v) b \mid \text{ctx}_v] = 0$. Taking outer expectation,

$$\mathbb{E}[\nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v) b] = 0. \quad (2)$$

Combining (2) with Assumption 1:

$$\begin{aligned} \mathbb{E}[\nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v) r_g] &= \mathbb{E}[\nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v) (r_g - b + b)] \\ &= \mathbb{E}[\nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v) (r_g - b)] + \mathbb{E}[\nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v) b] \\ &\stackrel{(1,2)}{=} 0 + 0 = 0 \quad \text{when measured w.r.t. the residual,} \end{aligned}$$

which means the advantage representation A_g may replace r_g in the estimator without changing the expectation, modulo the standardisation. The $\sigma_r + \delta$ scaling is a deterministic function of the batch and may be absorbed into a Hessian-free preconditioner; the μ_r subtraction is the classical GRPO baseline whose effect on the expectation is $O(1/G)$ via (2) applied to the dependent baseline (see Lemma 3 below). This establishes the per-node statement.

For the aggregate statement, write the expected reward functional as a sum over node contributions through the score-function identity:

$$\nabla_\theta J_{\mathcal{T}_g}(\theta) = \mathbb{E}\left[\sum_{v \in \mathcal{T}_g} \nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v) r_g\right].$$

With the $1/k_v$ weights, the conditional-expectation argument applied node-by-node yields

$$\mathbb{E}\left[\sum_v w_v \hat{g}_v\right] = \mathbb{E}\left[\sum_v w_v \nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v) r_g\right] + O(1/G),$$

which is the claim once one verifies that the w_v exactly compensate the expected branching factor (inductive argument on tree depth, omitted for brevity but identical to the depth-1 calculation in Section 2.4 of `02-math-deep-dive.md`). $\square$

Lemma 3 (Group baseline contributes $O(1/G)$ bias). *Let $\mu_r = \frac{1}{G} \sum_{g'} r_{g'}$ where r_g is one of the summed terms. Then $\text{Cov}(\nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v), \mu_r) = O(1/G)$.*

Proof. Only the g -th summand of μ_r depends on y_v ; the other $G - 1$ summands are independent draws by construction of the GRPO batch. Therefore the covariance reduces to $\frac{1}{G} \text{Cov}(\nabla_\theta \log \pi_\theta(y_v \mid \text{ctx}_v), r_g) = O(1/G)$. $\square$

Discussion

What Assumption 1 encodes. Equation (1) says: conditional on the context ctx_v that the node v inherits from its ancestors, the score function at v does not correlate with fluctuations in the root reward. This holds when the root reward is an *additive* function of independent child contributions — which is roughly the case for evidence-selection (the parent-rubric reward is roughly the sum of per-child snippet relevance scores, with each child responsible for a disjoint chunk).

What breaks the assumption.

- **Tool-use chains.** Child A’s output is the input to child B. The reward then has multiplicative dependence: a good A becomes wasted if B botches it, so the score function at B correlates with A’s output’s quality *through* the conditional reward.
- **Children with unrewarded sub-tasks.** If child C does bookkeeping with no contribution to r_g , the score at C is identically irrelevant to r_g — inheritance attaches reward noise to a useless gradient direction. Bias is zero but variance explodes.
- **Ancestor leakage.** If the same descendant prompt can occur under two different ancestor histories, then ctx_v may not be sufficient to absorb the conditional reward dependence.

Sharpening the open question. The blog flags “more fine-grained credit assignment can lead to faster convergence” as future work. Theorem 2 pins the exact failure mode: when (1) fails, the unbiased correction is to subtract from A_g the component of $r_g - \mathbb{E}[r_g \mid \text{ctx}_v]$ that the node v does *not* causally explain — i.e., a counterfactual or direct-effect estimator (e.g., REINFORCE with leave-one-out or causal-effect baselines).